a FlexLifeFi K-12 program

Ten Anchor Lessons

Standards, materials, procedure, differentiation - ready to teach.

Grade 9 year theme — Entrepreneurship Is Organized Problem-Solving.
The 9-12 band is built for the adults who will scrutinize: parents, teachers, administrators.

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GRADE 09 · LESSON PLANS

Ten Anchor Lesso

From flash of inspiration to repeatable discipline.

9

Ten fully fleshed lessons spanning the year. Each lesson is built to pass the picks-up-Monday test: a teacher with no prior prep should be able to read this Sunday night and teach it Monday morning. Standards are cited at the benchmark level (verbatim from the Standards Alignment Map). Differentiation notes are specific and actionable. Vocabulary is included for each lesson. Time blocks default to 55 minutes (single high-school period) unless noted; longer blocks are flagged.

WITH CASH - 9-12 VOICE

"What would have to be true for you to be wrong?"

Lessons are sampled across the year so the full arc is visible:

1. (Week 1) Cash V1 launch + The Riskiest Assumption
2. (Week 4) Mapping the Business Model Canvas — All Nine Boxes
3. (Week 10) The Mom Test — Interviewing Without Leading
4. (Week 14) Coding Five Interviews — From Transcripts to Hypothesis Revision
5. (Week 18) Customer Discovery Pack — Checkpoint 2 Studio Day
6. (Week 21) Building the One-Year P&L — From Canvas to Numbers
7. (Week 24) Compliance Is a Knowable List — License, Tax, ID, Age, PII
8. (Week 30) Designing the Two-Week MVP Test — "What Would Have to Be True?"
9. (Week 33) Conflict-Handling Styles — Pick the Move on Purpose
10. (Week 36) Final Capstone Defense — Kill, Iterate, or Scale

Lesson 1 — The Riskiest Assumption

> **"An MVP is the smallest test that could change your mind. It's not a v1."**
>
> - CASH, GRADE 9

Grade band: 9-12 **Time:** 55 minutes **Unit:** 1 — Map the model **Topic area:** Business Model Canvas launch · entrepreneurship as method

Objectives

By the end of this lesson, students will be able to:

- Write a "Day 1 Riskiest Assumption" for a hypothetical Capstone venture — the single assumption that, if wrong, would sink the whole thing.
- Distinguish a Canvas-shaped assumption (one of the nine BMC boxes) from a personal preference, with two examples.

Standards alignment

- **CEE Standard 1 (Earning Income), Grade 12:** Entrepreneurs accept the risks in organizing resources to produce goods and services and hope to earn profits.
- **CCSS-ELA W.9-10.4:** Produce clear and coherent writing in which the development, organization, and style are appropriate to task, purpose, and audience.
- **NGSS-ETS HS-ETS1-2:** Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems.
- **CASEL Responsible Decision-Making:** Analyze consequences of various actions.

Materials

- Cash Video #1 ("Entrepreneurship Is a Method, Not a Personality") — 2:50.
- Capstone Journal (composition notebook or digital equivalent) — one per student.
- Two index cards per student.
- Printed wall poster of the **Business Model Canvas** (9 boxes), enlarged enough to read across the room.
- A list of 10 sample assumptions (some Canvas-shaped, some preferences). Provided in program assets.

Vocabulary

- **Business Model Canvas (BMC)** — a 1-page map of a business across nine boxes: customers, value prop, channels, relationships, revenue, key activities, resources, partners, costs.
- **Hypothesis** — a claim you're not sure is true, written specifically enough to be checked.
- **Riskiest assumption** — the one assumption in your Canvas that, if wrong, would make the whole venture fail.
- **Assumption flag** — a written note next to a Canvas box that says "this is what I'm assuming; I have not yet checked it."

Procedure

1. ENGAGE — CASH VIDEO + ONE QUESTION (~8 MIN)

Play Cash Video #1. As students watch, post the prompt on the board:

> **"Think of a business idea you've had — even a half-formed one. What's the one thing about it that, if it turned out to be wrong, would mean the whole thing doesn't work?"**

Pair-share for 2 minutes. Cold-call three students to share. (You're surfacing a high-school version of the year's first muscle: distinguishing a risk you can name from a vague feeling that "it might not work.")

2. EXPLORE — CANVAS-SHAPED VS. PREFERENCE (~12 MIN)

Project the 10 sample assumptions, one at a time. For each, ask the class to call out Canvas-shaped or preference. Examples:

- "I think 16-year-olds will pay $15/month for a homework-help app." (**Canvas-shaped** — that's the customer + revenue assumption.)
- "I want my logo to be blue." (**Preference** — not a business risk.)
- "I'm assuming I can find a tutor partner who'll work for free." (**Canvas-shaped** — that's the key partner assumption.)
- "I think it'd be cool to have a podcast." (**Preference**.)
- "I'm assuming customers will check social media for their tutoring needs." (**Canvas-shaped** — that's the channel assumption.)

After about 7 statements, surface the rule: Canvas-shaped assumptions live inside one of the nine BMC boxes and would change the model if wrong. Preferences are about the founder, not the business.

3. EXPLAIN — THE NINE BOXES, BRIEFLY (~10 MIN)

Walk the wall poster of the BMC, naming each box in plain language:

1. **Customer segments** — Who, exactly, are you building for?
2. **Value propositions** — What problem are you solving for them?
3. **Channels** — How do they find you and get the product?
4. **Customer relationships** — What's the kind of relationship you're maintaining?
5. **Revenue streams** — How do you actually make money?
6. **Key resources** — What do you need to deliver this?
7. **Key activities** — What do you actually do all day?

8. **Key partners** — Who do you need on your side?
9. **Cost structure** — What does this cost you to run?

State: "By Week 8, you'll have all nine of these mapped — every box with either evidence or a flagged assumption." Add to Vocabulary Spine.

4. APPLY — DAY 1 RISKIEST ASSUMPTION (~17 MIN)

In the Capstone Journal, students write four things:

1. The kind of business they're imagining (one sentence).
2. **The Day 1 Riskiest Assumption.** Format: "This venture only works if [specific claim about specific people doing a specific thing]."
3. Which of the 9 Canvas boxes the assumption sits in.
4. One thing they would have to do this year to find out whether the assumption is true.

Confer with 5 students while others write — read their sentence, ask one sharpening question. ("That's about 'people' generically — which people? What age, where, doing what?")

5. CLOSE — PIN AND PREDICT (~8 MIN)

Each student writes the Day 1 Riskiest Assumption on an index card with their name. Collect them. Tell them: "You'll see this card again at three checkpoints — Weeks 8, 18, and 36. We're going to track whether the assumption holds, gets sharper, or breaks." Project this exit prompt:

> **"Predict: by Week 36, will this assumption (a) be confirmed, (b) be revised but largely intact, (c) break entirely, or (d) you don't know yet? Write a letter on the back of the card and one sentence why."**

Collect cards. End class.

Differentiation

- **ELL:** Provide the BMC poster pre-translated with cognates. Sentence frame: "This venture only works if [people] do [action] for [reason]." Allow native-language brainstorming with English finalization on the index card. Spanish cognates: segmento de clientes, propuesta de valor, canal, ingresos, costos.
- **IEP/504:** Allow a verbal version recorded on phone with student transcribing later. Provide a printed list of all 10 sample statements during the sort. Extended writing time on the Riskiest Assumption — okay to finish in the next session.

- **Advanced learners:** Push for two riskiest assumptions, one in customers/value prop (the demand side) and one in resources/partners/costs (the supply side). Have them rank: which would break the venture faster? This previews the unit-economics work in Unit 3.

Assessment

Formative: Watch the Canvas-shaped vs. preference call-and-response. Confer notes during writing time. **Summative:** Day 1 Riskiest Assumption is filed; it returns at Checkpoint 1 (Week 8) and is one of the year's reflection anchors.

Closure

Same as #5 above.

Homework / Extension

Read the 3-page summary of Business Model Generation (Osterwalder & Pigneur) in the Student Activity Book. Highlight one sentence about each of the nine boxes. Bring Friday.

Teacher notes

- The exercise of distinguishing Canvas-shaped from preference is the engine of this lesson. Don't skip it. Students who default to preference-thinking ("I want it to be cool") never get to evidence-thinking.
- Some students will struggle to name a Capstone topic on Day 1. That's fine — name a "kind of business" they're curious about. They'll lock the topic in Week 2.
- A student says "I don't have any ideas." Coach: "What's something you're already good at that someone else has paid for? What's something annoying that nobody fixes well?" Read the Skills + Hobbies + Needs framing if needed.

Lesson 2 — Mapping the Business Model Canvas — All Nine Boxes

Grade band: 9-12 **Time:** 90 minutes (block — or two consecutive periods) **Unit:** 1 — Map the model **Topic area:** Business Model Canvas · structured business modeling

Objectives

By the end of this lesson, students will be able to:

- Map all nine boxes of the Business Model Canvas for their Capstone venture, using either evidence (citing source) or a clearly flagged assumption.

- Identify which box currently holds the riskiest assumption and write a 2–3 sentence rationale.

Standards alignment

- **NGSS-ETS HS-ETS1-2:** Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems.
- **CCSS-ELA W.9-10.4:** Produce clear and coherent writing.
- **CEE Standard 1 (Earning Income), Grade 12:** Entrepreneurs accept the risks in organizing resources to produce goods and services and hope to earn profits.
- **C3 D2.Eco.2.9-12:** Use marginal benefits and marginal costs to construct an argument.

Materials

- BMC template (one per student, full-page; provided in assets).
- Sticky notes in 2 colors (green for evidence, yellow for flagged assumption).
- Capstone Journal.
- A worked-example BMC for a fictional venture ("Homework Helper" — a 1:1 tutoring service for 8th graders).
- Wall space for posting Canvases.

Vocabulary

- **Evidence** (recycled) — something you observed, heard, or measured.
- **Flagged assumption** — an entry in a Canvas box that you're labeling as not-yet-verified.
- **Customer segment** — a specific, identifiable group of customers, narrower than "everyone."
- **Value proposition** — what the customer gets that they couldn't get otherwise; the specific frustration you address.

Procedure

1. ENGAGE — WALKING THE WORKED EXAMPLE (~12 MIN)

Project the worked-example BMC for "Homework Helper." Read each box aloud. For each, identify whether it's evidence-backed or assumption-flagged. Examples:

- **Customer segment:** "8th-graders at Lincoln Middle School who are struggling with algebra." (Evidence: I asked the math teacher; she said 12 of her 28 students are failing the unit.)
- **Channels:** "Word of mouth at school + a flyer on the parent FB group." (Flagged assumption: I haven't tested the flyer yet.)
- **Revenue:** "$20 per session." (Flagged assumption: I picked the price; haven't asked anyone to pay it.)
- **Cost structure:** "$0/month — I tutor at home." (Evidence: confirmed with parents; no rent.)

Surface the rule: every box has a status. Either you have evidence, or you've flagged the assumption.

2. EXPLAIN — THE ORDER MATTERS (~8 MIN)

Walk through the order most students should fill the Canvas:

1. **Customer segment** first. (You can't write a value prop for "everyone.")
2. **Value proposition** second. (What frustration does this segment have that you address?)
3. **Channels and relationships** third. (How do you reach them? What kind of ongoing thing is this?)
4. **Revenue streams** fourth. (What's the actual money model?)
5. **Cost structure** fifth. (What does this cost to run?)
6. **Key activities, resources, partners** last. (What do you need on your side to make it happen?)

Note: Most beginners start with the product. Resist that. Customer first; product second.

3. APPLY — MAP YOUR OWN CANVAS (~50 MIN)

Students work independently on their own BMC. Process:

- Write the Canvas v1 entry in each box.
- Mark each entry green (evidence) or yellow (assumption) using sticky notes.
- For each yellow note, write a 1-sentence "what would tell me this is wrong?" on the back.

Confer with 8–10 students at 4 minutes each. Common stuck points and coaching:

- "I have no evidence for any box." Validate — that's normal at Week 4. Mark every box yellow. The point of the year is to turn yellow into green.
- "I have evidence for some, but it's all from family." That's an evidence type, but flag the bias. Note that interviews in Unit 2 will broaden the sample.
- "My value prop is 'a great experience.'" That's a preference, not a value prop. Push: "What specific frustration are you addressing? In what context?"

4. IDENTIFY THE RISKIEST BOX (~12 MIN)

Students step back and answer in the Capstone Journal:

- "My Canvas's riskiest box right now is [**_], because [**_]."
- "If I had to spend the next 4 weeks turning one yellow note green, it would be [____]."

Two volunteers share aloud. Class can react with one question each. Coach: the riskiest box is usually customer or value prop. If a student says "cost structure," push back — costs only matter if customers exist.

5. CLOSE — TAPE UP + ONE SENTENCE (~8 MIN)

Tape Canvases to the wall. Each student writes a 1-sentence "I notice..." on a sticky note about somebody else's Canvas (anonymously). Collect for return next class.

Differentiation

- **ELL:** Provide the BMC template pre-labeled with English + native-language box names. Sentence frames for each box ("My customer is ___ at **. They struggle with** ."). Pair with an ally for the green/yellow sort.
- **IEP/504:** Use a chunked Canvas — one box per page rather than a 9-box single-page version. Allow voice notes for each box, transcribed afterward. Reduce expected boxes-completed-in-class from 9 to 5 with the rest as homework.
- **Advanced learners:** Add a "second segment" Canvas — what if you targeted a different customer? Compare. Or push them toward a Lean Canvas variant (Maurya's adaptation, with "problem" and "unfair advantage" boxes) for an alternate framing.

Assessment

Formative: Walk during the mapping; check that students aren't writing "everyone" in customer segments. Read Capstone Journal entries during the riskiest-box step. **Summative:** The Canvas v1 is the Week 8 Capstone Pack core artifact, graded on the Business Plan Rubric (Rubric Bank).

Closure

Same as #5.

Homework / Extension

Capstone Journal: write a paragraph titled "Where my Canvas is riskiest, and what I'd ask a real customer to find out."

Teacher notes

- Canvases that are mostly yellow are not failing Canvases. They're honest. Reward that. The student who pretends to have evidence they don't have is the one to coach hard.
- A student writes "everyone" or "the world" in customer segment. That's the most common Week-4 mistake. Coach: "Pick a real human you could interview by Friday. Even just one."
- A student wants a perfect Canvas. There is no perfect Canvas. Coach: "Yours is supposed to change. The point of Unit 2 is to break and rebuild this."

Lesson 3 — The Mom Test: Interviewing Without Leading

Grade band: 9-12 **Time:** 55 minutes **Unit:** 2 — Discover the customer **Topic area:** Customer interviewing · evidence-gathering at high-school standard

Objectives

By the end of this lesson, students will be able to:

- Identify the three classic interview failure modes — compliments, hypotheticals, generic questions — using six example excerpts.
- Rewrite three "Mom-Test-fails" into Mom-Test-passes.

Standards alignment

- **CCSS-ELA W.9-10.7:** Conduct short and more sustained research projects to answer a question.
- **CCSS-ELA SL.9-10.1:** Initiate and participate effectively in collaborative discussions.
- **CCSS-Math S-IC.B.4:** Estimate from a sample.
- **CASEL Social Awareness:** Perspective-taking — letting the customer's view come through.

Materials

- The Mom Test 2-page summary (provided).
- 8 printed interview question cards (mix of Mom-Test-passes and -fails).
- Capstone Journal.
- Sample interview transcript A: a "complimentary" interview where the founder gets reassurance instead of evidence (1 page, provided).
- Sample interview transcript B: a "Mom-Test-passing" interview where the founder gets specific behavior data (1 page, provided).

Vocabulary

- **The Mom Test** — Rob Fitzpatrick's name for asking interview questions so good that even your mom couldn't lie to you. (The point isn't moms — it's avoiding questions that invite politeness.)
- **Compliment fail** — when the interviewee tells you your idea is great, and you treat that as evidence.
- **Hypothetical fail** — when you ask "would you buy X?" instead of "what did you do last time you faced this problem?"

- **Generic fail** — when you ask "do people like you usually...?" instead of "tell me about the last time you specifically did..."

Procedure

1. ENGAGE — TWO TRANSCRIPTS, SAME TOPIC (~10 MIN)

Project transcript A (compliment fail). Read aloud. Ask: "Would you start a business based on this?" Pair-share — most students say yes; the interviewee said the idea was great six times.

Project transcript B (Mom-Test pass). Read aloud. Same question. Pair-share — most students notice it's harder to read because the interviewee said specific things, some of which weren't flattering.

Surface: the first interview gave the founder a hug. The second gave the founder evidence. Only one of them is useful.

2. EXPLAIN — THE THREE FAILURES (~10 MIN)

Walk through:

- **Compliments.** "Your idea is great" is the lowest-information sentence in business. Ban it from your interview questions. Don't pitch the idea — ask about the customer's life.
- **Hypotheticals.** "Would you buy this?" is a coin flip; the customer is guessing. "Tell me about the last time you faced this problem — what did you do?" gets you behavior, which is the truth.
- **Generics.** "What do people like you usually do?" is fiction. "Walk me through your last week of trying to solve this" is a story.

Surface the rule: talk about the customer's life, not your idea.

3. APPLY — SORT AND REWRITE (~25 MIN)

Pass out the 8 question cards in pairs. Pairs sort each into Mom-Test-pass or Mom-Test-fail and tag the failure mode (C, H, G).

Sample cards:

- "Would you pay $15 for an app that helps you with homework?" (**H** — hypothetical)
- "Tell me about the last time you sat down to do homework — what made it hard?" (**Pass**)
- "Don't you think kids your age struggle with math?" (**C** — compliment-seeking, also leading)
- "Walk me through your study session yesterday — moment by moment." (**Pass**)
- "What do most teens do to handle homework stress?" (**G** — generic)
- "When was the last time you actually paid for something to help with school?" (**Pass**)
- "Would you tell your friends about this?" (**H**)

- "When did you last recommend something school-related to a friend? What did you say?" (**Pass**)

After the sort, each student picks 3 fails from the cards (or from the activity book's longer list) and rewrites each as a Mom-Test-pass. Then partner-checks.

4. CLOSE — BUILD TOMORROW'S INTERVIEW (~10 MIN)

Each student writes the first 5 questions of their own Capstone interview script. Rules: at least 4 of 5 must pass the Mom Test. No hypotheticals, no compliments, no generics. Confer briefly with as many students as possible. Collect at end.

Differentiation

- **ELL:** Provide a question stem bank ("Tell me about the last time you... What was annoying about... Walk me through... When did you last..."). Allow drafting in native language first; finalize in English. Pair with a strong English speaker for the partner check.
- **IEP/504:** Reduce rewrite count from 3 to 2. Provide pre-printed sentence stems. Allow audio recording of the 5 interview questions instead of writing.
- **Advanced learners:** Add a "follow-up question" requirement — for each Mom-Test-pass, write one follow-up that goes deeper into the story. Introduce probe and clarification as named moves. Pre-read excerpt from Steve Portigal's Interviewing Users if available.

Assessment

Formative: Sort accuracy in pairs. Watch the rewrites during conferring. **Summative:** The 5-question interview script feeds the interview activity in Weeks 11–13.

Closure

Same as #4.

Homework / Extension

Find a 5-minute YouTube interview with anyone (chef, athlete, founder). Transcribe 4 questions the interviewer asked. Tag each as Mom-Test-pass or -fail. Bring Wednesday.

Teacher notes

- The most common failure isn't a hypothetical — it's a flat-out pitch. A student says "my interview question is 'Would you buy a homework app that costs $15 per month and gives you 24/7 tutoring?'" That's not a question; it's a pitch. Coach: ask about their life, not about your idea.
- Some students will be nervous about interviewing. Validate, then practice with a peer first. Most kids fear it less after one round.

- The "mom" framing works at this age — sometimes too well. If a student is uncomfortable using a parent figure, swap to "the kindest neighbor on your block" or any non-confrontational person they know. The point is: don't ask questions that invite politeness over truth.

Lesson 4 — Coding Five Interviews: From Transcripts to Hypothesis Revision

Grade band: 9-12 **Time:** 90 minutes (block) **Unit:** 2 — Discover the customer **Topic area:** Qualitative coding · pattern recognition · hypothesis revision

Objectives

By the end of this lesson, students will be able to:

- Apply a 4-color coding system to 5 customer interview transcripts and tally the patterns.
- Write a 1-page "What changed in my hypothesis" memo identifying at least 2 specific Canvas changes driven by interview evidence.

Standards alignment

- **CCSS-Math S-IC.B.4:** Estimate from a sample.
- **CCSS-ELA W.9-10.7 / RST.9-10.7:** Conduct research and integrate technical information.
- **CCSS-ELA RI.9-10.1:** Cite strong and thorough textual evidence.
- **CASEL Self-Awareness:** Recognizing the gap between assumptions and evidence.

Materials

- Each student's 5 interview transcripts (collected over Weeks 11–13).
- Four highlighter colors per student (yellow = problem; blue = workaround; pink = surprise; green = direct quote worth keeping).
- Tally sheet (provided in activity book — 9-12 version with quote column).
- Each student's Canvas v1 (from Week 4 Lesson 2).
- Capstone Journal.

Vocabulary

- **Code** — a short label you put on a piece of evidence so you can count it.
- **Pattern** — something that shows up across at least 2 of 5 interviews.
- **Quiet signal** — something only one or two interviewees said, but that — if true — would matter a lot.

- **Hypothesis revision** — a change to your Canvas driven by interview evidence, with the change explicitly named.

Procedure

1. ENGAGE — SAMPLE CODED TRANSCRIPT (~10 MIN)

Project a sample 1-page transcript pre-coded by the teacher with all four colors. Walk it: yellow here is a frustration ("I waste 20 minutes every night just finding the right problem set"), blue here is a workaround ("she texts her older sister"), pink here is a surprise ("she said her phone is the problem, not the solution"), green here is a quote so good you'd want to put it in your Canvas memo verbatim.

Pose: "If you were the founder, which color would change your Canvas the most?"

Pair-share. Surface: yellow tells you the problem is real; pink often tells you you've been looking at the wrong problem.

2. EXPLORE — CODE YOUR FIVE (~30 MIN)

Students apply the 4-color code to all 5 of their own transcripts. Then they tally counts on the Tally Sheet, with the quote column filled for the 2–3 most powerful quotes per category.

Pace: about 6 minutes per transcript. Circulate. Most students will get through 4 in this block; finish the 5th in independent work time.

3. EXPLAIN — PATTERN, QUIET SIGNAL, ONE-OFF (~10 MIN)

Mini-lesson:

- A **pattern** shows up in at least 2 of 5 interviews. (At a sample of 5, you can't be statistically certain, but a 2-of-5 hit is meaningful.)
- A **quiet signal** is a 1-of-5 mention that, if true, would be high-impact. "One person said the price isn't the issue, the time commitment is. If that's true for more people, our pricing focus is wrong."
- A **one-off** is a single mention with low implied impact. Note it; don't act on it.

Make the rule explicit: "Treat patterns as evidence. Treat quiet signals as the next test. Treat one-offs as data you noted."

4. APPLY — WRITE THE HYPOTHESIS REVISION (~30 MIN)

In the Capstone Journal, students write a 1-page memo titled "What Changed in My Hypothesis." Required sections:

1. **My pattern (yellow):** "Across 5 interviews, [N] interviewees mentioned [problem]." Cite at least 2 quotes verbatim.

2. **My quiet signal:** "One interviewee said [_]. **If true, this would change [**_]." Defend whether to follow up.
3. **My biggest surprise:** "I expected [_]. **I learned [**_]. This makes me revise [____]."
4. **Changes to my Canvas:** "Box X — was [_], **now [**_], because [evidence]." At minimum 2 boxes.

Confer with as many students as possible. Push back on memos that say "nothing changed" — at this scale, something always changes, even if it's the founder's confidence in an assumption.

5. CLOSE — WALL OF REVISIONS (~10 MIN)

Each student writes their **biggest Canvas change** on a sticky note and posts to a class wall. Class walks the wall for 3 minutes. Two volunteers share what surprised them about the wall. End.

Differentiation

- **ELL:** Allow native-language interviews if customers were also ELL; transcripts can stay in original language with English summary required. Provide sentence frames for the 4-section memo. Pre-coded model transcript serves as a worked example.
- **IEP/504:** Reduce sample to 3 interviews coded in class; finish remaining 2 at home (or in resource room). Provide pre-printed Tally Sheet with the four colors labeled and quote columns. Audio submission of the memo allowed.
- **Advanced learners:** Push toward second-order coding — within yellow, are there sub-themes (price, time, quality, social)? Add a segmentation observation: "two of my five interviews look like one type; three look like another. Would I serve them differently?" This previews Grade 10 segmentation.

Assessment

Formative: Walk during coding; check that students aren't highlighting wholesale (every yellow defeats the purpose). Read memos during conferring. **Summative:** The hypothesis revision memo + coded transcripts feed the Customer Discovery Pack at Checkpoint 2 (Week 18), graded on the Customer Research rubric (9-12 band — full criteria).

Closure

Same as #5.

Homework / Extension

Bring Canvas v2 to Friday's class. Mark every box that changed since v1 with a red dot.

Teacher notes

- A student returns transcripts that are too short or unrevealing. Coach: "Which interview gave you the most? Code that one fully. Then come back to the others." Even a thin transcript yields something.
- Some students realize their interview script was bad (compliment-fishing, generics). Validate the realization — that's evidence too. Allow a 6th do-over interview if time permits.
- Watch for the "I confirmed everything I already thought" memo. That's almost never true at 5 interviews. Coach: "What did the customer say that you didn't expect? Even something small." The honest answer is gold.

Lesson 5 — Customer Discovery Pack Studio Day (Checkpoint 2)

> **"Don't make up a winner. Decide what to test next."**
>
> - CASH, GRADE 9

Grade band: 9-12 **Time:** 90 minutes (block) — Checkpoint day **Unit:** 2 — Discover the customer **Topic area:** Synthesis · positioning · public defense

Objectives

By the end of this lesson, students will be able to:

- Produce a Customer Discovery Pack consisting of: 5 coded transcripts, a hypothesis revision memo, a positioning statement, and a Canvas v2.
- Defend the Pack in a 5-minute peer + teacher panel review using at least 3 pieces of evidence.

Standards alignment

- **CCSS-ELA W.9-10.7 / RST.9-10.7:** Conduct research and integrate technical information.
- **CCSS-ELA SL.9-10.4:** Present information, findings, and supporting evidence clearly, concisely, and logically.
- **CCSS-Math S-IC.B.4:** Estimate from a sample.
- **NGSS-ETS HS-ETS1-2:** Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems.
- **CASEL Self-Awareness + Responsible Decision-Making.**

Materials

- Each student's 5 transcripts (coded), tally sheet, hypothesis revision memo, Canvas v2.
- Day 1 Riskiest Assumption index cards (returned from Lesson 1).
- Positioning statement template (provided).
- Customer Research rubric (Rubric Bank, 9-12 band).
- Wall space for posting Canvases v1 and v2 side by side.

Vocabulary

- **Positioning statement** — a 1-sentence claim of who the product is for, what it is, and why it's the right answer for that person. Form: "For [target] who [need], [product] is the [category] that [unique benefit] because [reason]."
- **Segmentation** — recognizing that "your customer" may actually be two distinct types of people who want different things.

Procedure

1. ENGAGE — HAND BACK DAY 1 CARDS (~5 MIN)

Hand back the Day 1 Riskiest Assumption cards. Quietly. Ask students to read their own and write one sentence in the Journal: "What's the biggest difference between this and what I now think?"

2. WRITE THE POSITIONING STATEMENT (~20 MIN)

Mini-lesson on the positioning statement format. Project:

> **"For [target] who [need], [product] is the [category] that [unique benefit] because [reason]."**

Worked example for the Homework Helper venture:

> **"For 8th-graders at Lincoln Middle School struggling with algebra, Homework Helper is the after-school tutoring service that delivers personalized 1:1 help in 30-minute sessions, because we use evidence from our own students' problem-set patterns to focus on the exact concepts each kid is missing."**

Walk the form: every blank must be specific (no "everyone," no "great experience"). Each blank should reference at least one piece of interview evidence.

Students draft, revise with a partner, and write the final.

3. BUILD CANVAS V2 (~25 MIN)

Students post Canvas v1 on the wall. Next to it, they post Canvas v2 — same template, but every change-since-v1 marked with a red dot. Each red dot has an evidence citation: which interview drove the change.

Confer with each student for 3–4 minutes.

4. PANEL DEFENSE (~30 MIN)

Form panels of 3 — student presenter, peer questioner, peer note-taker. Rotate so each student presents once and serves on two panels.

Presenter has 5 minutes:

- 1 minute Canvas v2 walk
- 1 minute riskiest assumption status (confirmed / revised / broken)
- 1 minute positioning statement read + defense
- 2 minutes patterns from interviews (with quotes)

Peer questioner asks 2 questions from this list:

- "Which of your 5 interviews surprised you most, and why?"
- "What would have to be true for your positioning statement to be wrong?"
- "What's the riskiest assumption that's still yellow on your Canvas?"
- "If you had to interview one more person before you'd believe your pattern, who would you pick and why?"

Peer note-taker scores against the Customer Research rubric (Rubric Bank, 9-12 — all 5 criteria).

5. SUBMIT (~10 MIN)

Customer Discovery Pack is submitted (digital or paper) for grading using the Customer Research rubric. Includes: 5 coded transcripts, tally sheet, hypothesis revision memo, positioning statement, Canvas v2.

Differentiation

- **ELL:** Provide the positioning template with each blank pre-explained and a vocabulary glossary on the back. Allow oral defense in pair-share if panel format is overwhelming. Permit translated quotes (with English summary) in the discovery pack.

- **IEP/504:** Pre-fill student name and date; chunk the positioning template across multiple lines. Extended time — finish the next class period. Allow audio submission of the memo and verbal panel defense.
- **Advanced learners:** Add a "secondary segment" Canvas — one you considered and rejected, with reasoning. Push for a unit-economics-aware positioning statement: not just "this is the category," but "this is why the math works at this segment."

Assessment

Summative: The Customer Discovery Pack is graded on the Customer Research rubric (9-12 band — 5+ interviews, formal coding, segmentation expected). Anchor papers from prior years (or instructor-built exemplars) are used to calibrate scoring.

Closure

Same as #5.

Homework / Extension

Capstone Journal entry: "What's one assumption still on my Canvas that I'd want to test in Unit 3 — and what would the test look like?"

Teacher notes

- A student's Day 1 assumption matches their Week 18 discovery. This is rare and usually means weak interviewing or a too-narrow sample. Probe: "What surprise did your interviews give you?" If they had no surprise, the interviews were probably leading.
- A student wants to use real classmates' names in their persona. Frame as fictional but plausible — the persona is a composite, not a person. Use initials in transcripts.
- The panel format is high-leverage; protect the time. Teach panel-question discipline ahead of time — peers are not here to "vote yes/no on the business," they're here to surface evidence gaps.

AN INTERLUDE · GRADE 09

"Which people, and how do you know?"

the program's spine question.

1

Lesson 6 — Building the One-Year P&L: From Canvas to Numbers

Grade band: 9-12 **Time:** 90 minutes (block) **Unit:** 3 — Model the money **Topic area:** Income statement modeling · unit economics

Objectives

By the end of this lesson, students will be able to:

- Build a 1-year P&L for their Capstone venture in a spreadsheet, including revenue, COGS, gross margin, operating expenses, and net income.
- Identify which 2 P&L lines are most sensitive to assumption changes and run one scenario (best/worst case).

Standards alignment

- **CCSS-Math A-CED.A.2:** Create equations in two or more variables.
- **CCSS-Math A-SSE.A.1:** Interpret expressions that represent a quantity in terms of its context.
- **Jump$tart Standard 1 (Spending and Saving), Benchmark 12-1:** Develop a plan for spending and saving — extended to a venture P&L.
- **CEE Standard 1 (Earning Income), Grade 12:** Entrepreneurs accept the risks in organizing resources to produce goods and services and hope to earn profits.
- **C3 D2.Eco.2.9-12:** Use marginal benefits and marginal costs to construct an argument.

Materials

- One-year P&L spreadsheet template (provided — Google Sheets / Excel for Web).
- Each student's Canvas v2 (from Checkpoint 2).
- Worked-example P&L for the "Homework Helper" venture.
- Capstone Journal.
- Calculator (or spreadsheet — at this band, spreadsheet is preferred).

Vocabulary

- **Revenue** — total sales over a period, before any costs are subtracted.
- **COGS (Cost of Goods Sold)** — variable cost of delivering each unit; for a service, the variable cost per session.
- **Gross margin** — Revenue – COGS, often expressed as a percent of revenue. The money left after the per-unit cost is paid.

- **Operating expenses (OpEx)** — fixed costs of running the business: subscriptions, advertising, fees, salary, etc.
- **Net income** — Gross margin − OpEx. The bottom line; what's left after everything.
- **Sensitivity** — how much net income changes when one input changes.

Procedure

1. ENGAGE — WALK THE WORKED EXAMPLE (~12 MIN)

Project the Homework Helper P&L. Read each line aloud, naming the assumption underneath:

- **Revenue:** $20/session × 4 sessions/week × 30 weeks = $2,400. Assumption: I get 4 paying sessions/week.
- **COGS:** $1.50 photocopies × 4/week × 30 = $180. Assumption: I print 4 problem sets/week.
- **Gross margin:** $2,400 − $180 = $2,220 (92.5%).
- **OpEx:** $0 advertising + $25/year (website domain) + $0 rent = $25.
- **Net income:** $2,220 − $25 = $2,195.

Surface: every line has an assumption underneath. The P&L is only as honest as those assumptions.

2. EXPLAIN — THE FIVE ROWS, IN ORDER (~10 MIN)

Walk through the structure:

1. **Revenue.** Price × volume. Each is its own assumption.
2. **COGS.** Per-unit variable cost × volume. Recycles Grade 7 per-customer math.
3. **Gross margin.** = Revenue − COGS. The "what's left after making the thing" number.
4. **Operating expenses.** Fixed costs. Subscriptions, advertising, fees, etc.
5. **Net income.** = Gross margin − OpEx.

Surface the rule: gross margin pays for OpEx and profit. If gross margin can't cover OpEx, you don't have a business — you have a hobby with bills.

3. APPLY — BUILD YOUR P&L (~45 MIN)

Students open the spreadsheet template and build their Capstone's 1-year P&L. Process:

- Fill in revenue first (price × volume; both assumptions).
- COGS next (variable cost × volume).
- Gross margin (formula auto-calculates).
- OpEx (list each line item).
- Net income (formula auto-calculates).

Confer with 8–10 students at 4 minutes each. Common stuck points:

- "I have no costs." Almost never true. Coach: "What does it cost you per unit to deliver? Even if it's just printing or your phone data?"
- "My revenue line is just one number." Coach: "Show the math. Price × volume."
- "I have no idea what my volume will be." That's fine — write the assumption and what you're basing it on. Even "5 customers/week because I have 5 friends who said they'd be interested" is a defensible starting point. Note the bias.

4. RUN A SCENARIO (~15 MIN)

Each student runs one scenario test:

- Identify which 2 inputs are most uncertain (usually price and volume).
- Build a "worst case" column: drop both by 30%.
- Build a "best case" column: raise both by 30%.
- See what happens to net income.

Coach: "Where does your P&L break? What volume of sales would have to be true for you to be net-income-positive?" This is break-even analysis — recycled from Grade 8 — now embedded in a P&L.

5. CLOSE — RISKIEST NUMBER (~8 MIN)

Capstone Journal:

- "The riskiest number on my P&L is [**_], because if it's wrong by [**], the bottom line goes from [**_] to [**]."
- "To check this number, I would [____]."

End class.

Differentiation

- **ELL:** Pre-translate the spreadsheet template's row labels. Cognates in Spanish: ingresos (revenue), costos (costs), margen (margin), ingresos netos (net income). Pair on the build for the first 15 minutes.
- **IEP/504:** Provide a paper version of the P&L with rows pre-labeled and formulas explained. Calculator allowed throughout. Reduce scenario count from 2 (best/worst) to 1 (worst case only).
- **Advanced learners:** Add a "what's my contribution margin per customer" row above gross margin. Compute breakeven volume (fixed costs ÷ contribution margin). Have them compare with their stated volume assumption — are they planning to operate above or below breakeven?

Assessment

Formative: Walk during the build; check that revenue is computed (not stated). Read the Riskiest Number journal entries during conferring. **Summative:** The 1-year P&L is the core artifact in the Money Model Pack at Checkpoint 3 (Week 28), graded on the Budget Project Rubric (9-12 band — full complexity).

Closure

Same as #5.

Homework / Extension

Capstone Journal: "If I had to actually start this venture next month, what's the smallest amount of money I'd need? What is that money for?" — this previews Grade 10's funding-source comparison.

Teacher notes

- Watch for the "I'll figure it out" line. There's no "figure it out" line in a P&L. Every line is a number. Coach toward stating the number — even badly — and flagging it as an assumption.
- A student's P&L shows them losing money. That's not a failed P&L that's a useful one. Coach: "What would have to change to flip this? Higher price? Lower COGS? More volume? What would each cost in effort?"
- A student's P&L shows them making $100,000 in year 1. Push back hard. "What's the volume assumption? Do you have evidence that supports that volume?" The honest P&L is rarely a hockey-stick.

Lesson 7 — Compliance Is a Knowable List

Grade band: 9-12 **Time:** 55 minutes **Unit:** 3 — Model the money **Topic area:** Ethics, safety, legal awareness · compliance basics

Objectives

By the end of this lesson, students will be able to:

- Identify which of five compliance items applies to their Capstone venture: business license, sales tax, ID-required products, age limits on platforms, PII protection.
- Write a defined response for each applicable item.

Standards alignment

- **Jump$tart Standard 3 (Employment and Income), Benchmark 12-3:** Explain how taxes, government transfer payments, and employee benefits relate to disposable income.
- **CCSS-ELA W.9-10.4:** Produce clear and coherent writing.
- **C3 D2.Eco.2.9-12:** Use marginal benefits and marginal costs to construct an argument.
- **CASEL Responsible Decision-Making:** Considering ethical responsibility.

Materials

- Compliance checklist template (5 rows: license, tax, ID, age, PII; columns: applies to my venture? · what to do · who to ask · status).
- One short anonymized case study of a teen-run venture that hit a compliance bump (provided in assets — picked from common patterns: retroactive sales tax, age-limit account ban, PII leak).
- Each student's Canvas v2 + P&L draft.
- Capstone Journal.
- Pre-pinned state-specific links (teacher does this once per state — small business administration, state tax authority, FTC kids privacy basics).

Vocabulary

- **Business license** — a permit your state, county, or city requires before a business operates legally. Often free or low-cost for small ventures.
- **Sales tax** — the percent your state collects on most retail transactions; you're responsible for collecting it from customers and remitting it to the state.
- **ID-required product** — products that are illegal to sell to minors (alcohol, tobacco, certain supplements) — relevant if any teammate is selling/buying these.
- **Age limit** — most major platforms (most social media, most marketplaces) require account holders to be 13+ (some 18+). Violating is a real risk.
- **PII (Personally Identifiable Information)** — information that identifies an individual: full name, address, phone, email, photo, school, financial details. Recycled from Grade 8 with extension.

Procedure

1. ENGAGE — THE COMPLIANCE BUMP CASE (~10 MIN)

Read the anonymized case aloud (1–2 pages). Suggested case: a 16-year-old running a custom-art Etsy shop who didn't register for sales tax, sold $1,400 in a year, and got a letter from the state tax authority asking for back taxes and a small penalty.

Pose: "Was she doing anything wrong? What could she have done up front?"

Pair-share. Surface: she wasn't wrong morally; she was wrong administratively. Compliance isn't ethics — it's logistics. The fix is the checklist.

2. EXPLAIN — THE FIVE ITEMS (~12 MIN)

Walk through each item, using a worked example from a hypothetical food-service venture (a teen running a weekend baked-goods stand):

1. **Business license.** Applies? Often yes for food sales. What to do? Check your city/county clerk's office; many issue a "cottage food" or "home-based business" permit free or for $25. Who to ask? The clerk's office; the state's small business administration website.
2. **Sales tax.** Applies? In most states, yes for tangible goods. What to do? Register for a seller's permit (often free); collect tax at point of sale; remit quarterly or annually. Who to ask? State tax authority.
3. **ID-required products.** Applies? Not for baked goods. (But would for, say, energy drinks with kratom or CBD — flag if your student is selling anything in this gray area.)
4. **Age limit.** Applies? If selling on a marketplace (Etsy, eBay, social media), check the platform's age policy. Most require 13+ for accounts; some require 18+. Some require parent-supervised accounts under 18. What to do? Read the ToS; involve a parent-supervised account if appropriate.
5. **PII protection.** Applies? Always. What to do? Use initials in customer transcripts, don't post your home address as the business address (use a P.O. box, parent permission, or school address with permission), don't share customer data publicly without consent.

Surface the year's frame: "Compliance isn't the wall. It's the doorway."

3. APPLY — FILL YOUR CHECKLIST (~25 MIN)

Students complete the compliance checklist for their own Capstone:

- For each of 5 rows, mark Applies (Y/N).
- For Y rows, write the defined response in the "what to do" column.
- For Y rows, write who they'd ask to verify.
- Mark current status: Done · In Progress · Not Yet Started.

Confer with 8–10 students. Common stuck points:

- "I'm just doing this for class — does it apply?" Coach: complete the checklist as if it were real. The exercise is the learning.
- "I don't know if my state requires X." Use the pre-pinned state links. The point is learning to look it up, not memorizing it.

4. CLOSE — ONE SENTENCE RESPONSE (~8 MIN)

Capstone Journal:

- "The compliance item I take most seriously for my venture is [**_], because [**_]."
- "The one I'm least sure about is [**_]; I'd find out by [**_]."

End. Two volunteers share aloud.

Differentiation

- **ELL:** Provide the checklist template pre-translated. Cognates: licencia, impuesto sobre ventas, edad límite, datos personales (PII). Allow native-language outreach to family for the "who to ask" column if a parent or relative runs a small business.
- **IEP/504:** Reduce checklist from 5 to 3 most-relevant items. Pre-fill the "who to ask" column with default sources. Audio submission of the journal entry.
- **Advanced learners:** Add a sixth row — contracts and terms of service. What does a 1-page customer agreement need to say? Optional Grade 10 preview: introduce sole prop vs. LLC briefly, with one-line explanations.

Assessment

Formative: Walk during the checklist completion. Read journal entries. **Summative:** The compliance checklist is part of the Money Model Pack at Checkpoint 3.

Closure

Same as #4.

Homework / Extension

Pick one compliance item that applies. Write the actual email or message you'd send to ask the right office (tax authority, city clerk, platform support). Bring Friday.

Teacher notes

- Compliance can scare students into not doing the venture. Counter-frame: most teen ventures clear compliance with about an hour of work and zero dollars. The fear is bigger than the work.
- A student's venture has a real legal issue (selling restricted items, planning to use a fake age on a platform). This is a teaching moment — coach toward a pivot. The optional pivot window in Week 19 is for this.
- **Sensitivity flag:** The PII conversation can surface that a student already has personal details (full name + photo + address) on a public platform. Pause. Coach toward a parent conversation. Direct the family to the Parent Guide.

Lesson 8 — Designing the Two-Week MVP Test: "What Would Have to Be True?"

Grade band: 9-12 **Time:** 90 minutes (block) **Unit:** 4 — Test the bet **Topic area:** MVP design · falsifiable hypothesis · marginal-benefit reasoning

Objectives

By the end of this lesson, students will be able to:

- Design a 2-week MVP test with a defined "what would have to be true" success metric.
- Write a kill / iterate / scale recommendation rule before the test runs (so the test result is interpretable).

Standards alignment

- **NGSS-ETS HS-ETS1-2:** Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems.
- **C3 D2.Eco.2.9-12:** Use marginal benefits and marginal costs to construct an argument.
- **CCSS-Math S-IC.B.4:** Estimate from a sample.
- **CCSS-ELA W.9-10.7:** Conduct research projects.

Materials

- 2-week MVP test plan template (provided).
- Cash Video #7 ("The Smallest Test That Could Change Your Mind") — 2:50.
- Each student's Canvas v2, P&L, and Compliance checklist.
- Capstone Journal.
- One worked example: the Warby Parker home-try-on test (5 customers, 5 frames, mailed back).

Vocabulary

- **MVP (Minimum Viable Product)** — the smallest test that could change your mind about whether to continue.
- **What-would-have-to-be-true (WWHTBT)** — a pre-committed success metric. "For me to keep going, I would need to see X."
- **Kill** — the test result tells you the assumption is wrong; you stop or pivot.
- **Iterate** — the test result is mixed; you refine and test again.
- **Scale** — the test result confirms the assumption strongly enough that you'd put more resources in.

- **Premortem** — imagining the test failed and listing the reasons why, before it runs. (Light introduction; full treatment in Grade 10.)

Procedure

1. ENGAGE — CASH VIDEO + WARBY PARKER (~12 MIN)

Play Cash Video #7. Then walk the Warby Parker case in 3 minutes:

- The founders weren't sure if people would buy glasses online without trying them on first.
- Their MVP wasn't a website. It wasn't a store. It was a 5-frame home try-on box, sent free, with prepaid return.
- They mailed 5 boxes. Tracked who tried what, who bought, who returned without buying.
- Their what-would-have-to-be-true was: at least 2 of 5 customers buy after trying. It was simple. It was checkable.

Pose: "What was the smallest version of your venture that would tell you whether to keep going?"

Pair-share. Cold-call three students.

2. EXPLAIN — THE THREE PARTS OF A GOOD MVP TEST (~12 MIN)

Walk through:

1. **One assumption to test.** Pick from your Canvas. Usually customer or value prop. "My assumption is that 8th-graders will pay $20 for a 30-minute tutoring session."
2. **A what-would-have-to-be-true metric.** Specific, pre-committed. "For me to continue, I'd need at least 4 of 5 outreach attempts to result in a paid session within 2 weeks."
3. **A kill / iterate / scale rule.** "If 0–1 sessions: kill or pivot. 2–3: iterate (different price or messaging). 4–5: scale (try 10 more outreaches)."

Surface the rule: if you can't say what result would change your mind, your test isn't designed yet.

3. APPLY — DESIGN YOUR 2-WEEK TEST (~45 MIN)

Students fill in the 2-week MVP test plan template. Required sections:

- **Assumption being tested.** (Cite which Canvas box.)
- **What I'm doing.** (The actual activity. Mailing 5 home-try-ons. Posting in 3 parent FB groups and tracking response. Doing 4 free trial sessions and converting at the end.)
- **What-would-have-to-be-true metric.** (One specific number.)
- **Kill / iterate / scale rule.** (Pre-committed.)
- **Compliance check.** (Cross-reference Lesson 7 — does this test require any compliance moves?)

- **Premortem (light).** (3 reasons this test could fail that would invalidate the data, not just the venture. Example: "It's spring break so volume is artificially low.")
- **Timeline.** Week 1 actions, Week 2 actions, decision day.

Confer with 8–10 students at 4 minutes each.

4. MARGINAL-BENEFIT CHECK (~12 MIN)

Walk through the marginal-benefit/marginal-cost frame (per the C3 standard):

- Marginal benefit of running the test: clearer evidence about my riskiest assumption.
- Marginal cost of running the test: the time, money, and emotional risk involved.
- If marginal benefit > marginal cost: run the test. If not: redesign or skip.

Each student writes a 1-paragraph "Should I run this test?" decision in the Journal. Coach: if the test would cost you 10 hours and tell you something you already know, redesign.

5. CLOSE — SIGN-OFF (~10 MIN)

Each student gets a partner-check. The partner reads the test plan and signs off (or doesn't) using these questions:

- Is the assumption clearly named?
- Is the WWHTBT metric specific enough that you'll know who won?
- Is the kill/iterate/scale rule written before the test runs?

If any "no," revise. Collect signed plans at end.

Differentiation

- **ELL:** Provide the test plan template with pre-translated section headers. Sentence frames for the WWHTBT metric: "For me to continue, I would need ___ to ___ within ___."
- **IEP/504:** Reduce premortem reason count from 3 to 1. Allow audio submission of the test plan.
- **Advanced learners:** Add a "what would tell you the test itself failed (vs. the assumption failed)" section. This is a sophisticated move that distinguishes founders from beginners — knowing whether the result invalidates the bet, or whether the test was contaminated.

Assessment

Formative: Confer during design. Sign-off step. **Summative:** The MVP test plan is the spine of the MVP Test Pack at Checkpoint 4 (Week 36).

Closure

Same as #5.

Homework / Extension

Send the first outreach (or take the first action) by Wednesday. Capstone Journal: "What I did, what happened, what surprised me."

Teacher notes

- A student's WWHTBT metric is too vague ("a lot of people"). Coach: "Pick a number. Five of ten. Two of three. Be wrong on the number if you have to — but pick one. The whole point is committing before the result comes in."
- A student wants to skip the kill/iterate/scale rule. Don't let them. The rule is the test's most important sentence — without it, students will rationalize whatever happened as "kind of working."
- A student's MVP is too big (a full-feature website, a 6-month plan). Coach toward smaller. "What's the smallest version of this that would tell you the same thing?" Often the answer is: a 5-customer email, a flyer in 3 places, one free trial.

Lesson 9 — Conflict-Handling Styles: Pick the Move on Purpose

> **"Revenue is loud. Contribution margin is true. Compute one number this week."**
>
> - CASH, GRADE 9

Grade band: 9-12 **Time:** 55 minutes **Unit:** 4 — Test the bet **Topic area:** Leadership · self-awareness · team dynamics

Objectives

By the end of this lesson, students will be able to:

- Identify their own default conflict-handling style using a 12-question self-inventory across the 5 styles: avoid, accommodate, compete, compromise, collaborate.
- Pick one style they want to improve and write a 3-step practice plan.

Standards alignment

- **CASEL Relationship Skills:** Communication, conflict management.
- **CASEL Self-Awareness:** Identifying personal patterns.

- **CCSS-ELA W.9-10.4:** Produce clear and coherent writing.
- **CEE Standard 1 (Earning Income), Grade 12:** Entrepreneurs accept the risks in organizing resources to produce goods and services and hope to earn profits. (Applied: people-management is part of running a venture.)

Materials

- Conflict-handling style self-inventory (12 items, Likert scale, adapted from Thomas-Kilmann). Provided in assets.
- One-page reference: the 5 styles + when each is useful.
- Capstone Journal.
- Three short 1-page case scenarios (a teammate is missing deadlines · a peer wants to change the venture mid-test · a partner makes a unilateral pricing decision). Provided in assets.

Vocabulary

- **Avoid** — sidestepping or postponing the conflict. Useful when issue is trivial or emotions are too high to discuss productively.
- **Accommodate** — letting the other person have what they want. Useful when the issue matters more to them, or when preserving the relationship is most important.
- **Compete** — pushing for your position. Useful when the issue is critical, evidence is on your side, and time is short.
- **Compromise** — both sides give something up. Useful when the stakes are moderate and a quick decision is needed.
- **Collaborate** — working together to find a solution that meets both interests. Useful when the issue matters to both, time exists, and trust is present.

Procedure

1. ENGAGE — A STORY FROM A REAL TEAM (~7 MIN)

Tell a brief 3-minute story (made up to be true): a Capstone team last year had a conflict over whether to pivot the venture in Week 20. One student's interview evidence said pivot. Another's said stay. One avoided the conversation for two weeks. One fought hard for staying. One proposed running both versions in parallel. The team that survived was the one that named the conflict — and named which style each person was using.

Pose: "Which person was right?" Pair-share. Surface: the question isn't who's right. It's what each person did with their position.

2. EXPLAIN — THE FIVE STYLES (~10 MIN)

Walk through the styles. Display a 2×2 grid: cooperativeness (low/high) on one axis, assertiveness (low/high) on the other.

- **Avoid:** low cooperativeness, low assertiveness.
- **Accommodate:** high cooperativeness, low assertiveness.
- **Compete:** low cooperativeness, high assertiveness.
- **Compromise:** medium on both.
- **Collaborate:** high on both.

Surface the rule: no style is "good" or "bad." Each is right somewhere. The skill is choosing on purpose.

3. APPLY — TAKE THE INVENTORY (~12 MIN)

Students complete the 12-item self-inventory individually. Tally to identify default style. Then read the corresponding "when this style helps / when this style hurts" page in the activity book.

4. APPLY — THREE CASE SCENARIOS (~15 MIN)

In groups of 3, students read the three case scenarios. For each, they discuss:

- What's the appropriate style here?
- What's the wrong style — the one a beginner might default to?
- What would they actually do if it were them?

Share-out: 1 case, 2 groups share.

5. CLOSE — ONE STYLE TO IMPROVE (~8 MIN)

Capstone Journal:

- "My default conflict-handling style appears to be [____]."
- "One style I want to improve is [**_], because [**_]."
- "Three small ways I'll practice this in the next four weeks: [**_], [**_], [____]."

End. One volunteer shares.

Differentiation

- **ELL:** Provide the self-inventory pre-translated; cognates: evitar, acomodar, competir, comprometer, colaborar. Pair on case discussion with an ally.
- **IEP/504:** Reduce inventory from 12 items to 8. Permit verbal sharing of the case discussion instead of written. Provide style-summary cards for reference.

- **Advanced learners:** Add a fourth scenario — a conflict with an adult mentor or family member about the venture (continuing or pivoting). The adult-conflict scenario adds complexity around power dynamics.

Assessment

Formative: Read Capstone Journal entries. **Summative:** The conflict-style identification + practice plan is part of the team retrospective in the Final MVP Test Pack at Checkpoint 4.

Closure

Same as #5.

Homework / Extension

Notice one conflict (small or large) in the next 4 days. Capstone Journal: which style did you default to? Was it the right one? What would you have tried instead?

Teacher notes

- The inventory is self-perception, not diagnosis. Coach: "Your style isn't fixed. The point of naming it is to choose it on purpose next time."
- A student says "I never have conflict." Probe: are they avoiding, or accommodating, or just lucky? Often the "no conflict" student is the one accommodating most.
- **Sensitivity flag:** the conflict-style frame can land near home dynamics. If a student starts disclosing family conflict in detail, listen briefly, validate, and route to school counselor if the issue is big. Don't try to handle it inside the lesson.

Lesson 10 — Final Capstone Defense: Kill, Iterate, or Scale

Grade band: 9-12 **Time:** 100 minutes (block — or two periods) **Unit:** Capstone synthesis
Topic area: Reflection · public defense · year-long synthesis

Objectives

By the end of this lesson, students will be able to:

- Deliver a 7-minute Capstone Defense covering: (a) Canvas v2 + riskiest assumption status, (b) customer discovery findings, (c) 1-year P&L key numbers, (d) 2-week MVP test results + kill/iterate/scale recommendation, (e) personal reflection (what changed, conflict-style learning).

- Respond to 4 panel questions citing specific evidence from the year's work, including at least one evidence-portfolio reference.

Standards alignment

- **CCSS-ELA SL.9-10.4:** Present information, findings, and supporting evidence clearly, concisely, and logically.
- **CCSS-ELA SL.9-10.5:** Make strategic use of digital media in presentations.
- **CCSS-ELA W.9-10.7 / RST.9-10.7:** Conduct research and integrate technical information.
- **NGSS-ETS HS-ETS1-2:** Design a solution to a complex real-world problem.
- **CASEL Self-Awareness + Responsible Decision-Making.**

Materials

- Each student's full Capstone Journal + portfolio (Canvas v1 and v2, 5 transcripts, hypothesis revision memo, positioning, P&L, compliance checklist, MVP test plan + results).
- Defense slides or 1-page handout (built in Week 35).
- Pitch Presentation Rubric (Rubric Bank, 9-12 band — full Q&A criterion).
- Panel of 3–5 evaluators (other teachers, parents, older students, community members, ideally including one teacher from Grade 8 and one from Grade 10 for vertical handoff).
- Timer.
- Day 1 Riskiest Assumption cards (returned at end).

Vocabulary

(All previously introduced — recycle.)

Procedure

1. ENGAGE — THE 90-SECOND WARM-UP (~12 MIN)

Pair students up. Each gives a 90-second "elevator" version of their Capstone to their partner — Canvas, customer, P&L, MVP, recommendation. Partner asks one question. Repeat with a new partner. (Two rounds total. This is a confidence warm-up before the panel.)

2. SET THE NORMS (~5 MIN)

State the day's norms clearly:

- Each student gets 7 minutes for the defense + 5 minutes for panel Q&A.
- The panel is curious, not adversarial. They look for evidence and reflection, not for "is this a great business?"
- The audience listens silently and writes one "I noticed..." and one "I wondered..." per presenter.

- The "honest claim" criterion is non-negotiable. A polished defense built on a misleading number scores low on that row no matter how smooth.

Show the rubric (Pitch Presentation, 9-12) one more time.

3. DEFENSES (~70 MIN, ~10–12 STUDENTS AT 12 MIN EACH; LONGER BLOCK OR SPLIT DAY FOR MORE)

Each student delivers the 7-minute defense with the structure:

- **The Canvas (1 min):** "My venture is X. My riskiest Day-1 assumption was Y. Today, that assumption status is [confirmed / revised / broken] because..."
- **The customer (1.5 min):** "I learned my customer is Z, and here's what's specific about them — quote, quote, quote. The biggest thing I got wrong on Day 1 was..."
- **The math (1.5 min):** "My 1-year P&L. Revenue $**, COGS $**, gross margin $**, OpEx $**, net income $___. The riskiest number is ___ because..."
- **The MVP test (2 min):** "My WWHTBT was **. Result was** . My pre-committed kill/iterate/scale rule said **. So my recommendation is** ."
- **What I learned about myself (1 min):** "My default conflict-handling style is **. The biggest thing I'd do differently next year is** ."

After each: 5 minutes of panel Q&A. Timer keeper enforced strictly.

Panel question bank (provide to panel at briefing):

- "Walk me through one quote from your interviews that changed your Canvas. Why?"
- "What's the assumption in your P&L you're least sure about, and how would you check it?"
- "If your MVP came back the opposite of what it did, what would you do?"
- "Where did your compliance checklist matter? Where did it not?"
- "What's the conflict-style move you defaulted to this year that you wish you hadn't?"

4. AUDIENCE REFLECTION (~5 MIN)

Each audience member completes a 2-question card:

- One thing I noticed across many defenses today: __.
- One thing I'd want to learn more about: __.

Collect.

5. FINAL CASH REFLECTION (~10 MIN)

Play Cash Video #8 — the year-end "Organized Problem-Solving" summary. Students write one final Capstone Journal entry:

- The biggest thing I learned this year was [____].
- One thing I'd tell next year's Grade 9 student is [____].

- The Day 1 Riskiest Assumption ended up [____]. Here's what I now think.
- I'm proud of [____].
- I'm going to keep working on [____].

6. CLOSE — DAY 1 CARD RETURN (~8 MIN)

Hand back Day 1 Riskiest Assumption cards. Students re-read. Two volunteers share what 36 weeks of evidence did to it.

Differentiation

- **ELL:** Allow visuals and a 1-page handout to carry more weight in the defense — Canvas, P&L, MVP results in chart form. Sentence frames for Q&A: "Based on [evidence], I think [_] **because [_]**." Permit a same-language peer to translate panel questions on the spot if needed.
- **IEP/504:** Allow the defense to be delivered in shorter chunks if 7 minutes is overwhelming — 1-minute mini-segments with a 30-second pause. Permit a printed script. Audio-record version submitted in advance is also acceptable. Extended Q&A response time.
- **Advanced learners:** Add a "what would I do with $1,000 to run this for real next month" segment. Push for a written 1-page addendum identifying the highest-leverage next step. This previews Grade 10 ownership-and-funding work.

Assessment

Summative: The defense is graded on the Pitch Presentation rubric (9-12 — full Q&A criterion). Capstone Journal across the full year is a separate portfolio grade using the Reflection Writing rubric.

Closure

Same as #6.

Homework / Extension

(End of year — none required.) Optional: send the MVP test summary + recommendation to the Grade 10 teacher for a vertical handoff conversation in fall.

Teacher notes

- Panel members need a 15-minute briefing before defenses start. Cover: rubric, norms, what to ask for evidence, what not to ask (no judgments on whether the venture would "really work" in market). Provide the question bank above.
- Some students will be very nervous. The 90-second warm-up matters. So does the audience-card structure — it makes peers active listeners, not silent watchers.

- The Day 1 card return at the end is the year's emotional payoff. Don't skip. Some students will laugh at how wrong they were on Day 1; some will be moved by how much they've learned. Both are the right response.
- If a student's recommendation is "kill," celebrate the discipline. The student who kills a venture honestly outscores the student who claims to scale a venture they have no evidence for.

AN INTERLUDE · GRADE 09

"Most adults skip the step. You won't."

the band's working assumption.

2

Senior expert upgrade pack — Grade 9 worked examples, misconceptions, and contracts

Three full P&L worked examples (deepening Lesson 6)

The original lesson teaches the form. Add three end-to-end P&Ls at three different revenue scales.

WORKED EXAMPLE A — A SIDE-HUSTLE SERVICE (ANNUAL P&L)

Lawn-mowing service, 1 mower, 1 student.

Line	Year 1
Revenue (40 lawns × $40 avg)	$1,600
COGS (gas, blade sharpening, $5/lawn)	$200
Gross margin	**$1,400 (87.5%)**
Operating expenses (flyers, tools, app)	$200
Net income (pre-tax)	**$1,200**

Coach: "This is small but real. Margin is high because the service is mostly time. The business mostly turns hours into dollars."

WORKED EXAMPLE B — A SMALL PRODUCT BUSINESS (ANNUAL P&L)

Custom T-shirt brand, 200 shirts/year.

Line	Year 1
Revenue (200 × $25 avg)	$5,000
COGS (blank shirt + printing, $10/shirt)	$2,000
Gross margin	**$3,000 (60%)**
Operating expenses (website, marketing, supplies)	$1,200
Net income (pre-tax)	**$1,800**

Coach: "Lower margin than the service business. Volume is the lever. To double net income, you don't need to double revenue — you need to double volume."

WORKED EXAMPLE C — A SUBSCRIPTION CONTENT BUSINESS (ANNUAL P&L)

Newsletter, 100 paid subscribers at $5/mo.

Line	Year 1
Revenue (100 × $5 × 12 mo)	$6,000
COGS (email service, hosting)	$480
Gross margin	**$5,520 (92%)**
Operating expenses (research tools, design)	$600
Net income (pre-tax)	**$4,920**

Coach: "Highest margin, lowest variable cost. But the customer-acquisition challenge is the real cost — implicit, not on this P&L. Discuss: where would you put 'time spent on content'?" Surface labor as opportunity cost; not on the P&L but real.

Common Misconceptions (Grade 9)

Belief	Reality	Coach
"MVP means launch-ready."	MVP means "the smallest version that could change my mind." It's an experiment, not a release.	"What would the test cost you? What would the test tell you? If the answer to either is 'I don't know,' it's not an MVP yet."
"Every BMC box needs to be perfect on Day 1."	Every box starts as an assumption. Half are wrong. The work is finding which ones.	"Mark each box with a flag: confirmed, hypothesis, or imagined. Most should start as imagined."
"If I do customer discovery, I'll find the truth."	Customer discovery surfaces patterns, not absolutes. 5 interviews is enough to disconfirm a strong assumption, not enough to declare a winner.	"What did the interviews change in your hypothesis? If nothing changed, your interviews were probably leading."
"P&L tells me if my business is healthy."	P&L tells you if the math works. Cash flow tells you if you stay alive month to month. Both matter. (Cash flow is Grade 11.)	"Profitable on paper, broke in the bank. Both are possible. We'll fix that next year."
"Customer interviews are sales pitches."	Customer interviews are listening. The Mom Test rule: ask about their life, not your idea.	"Did you describe your idea? If yes, the interview was contaminated. Ask about their problem next time."

Half-lesson — Contracts basics for a kid-run service

Time: 30 min (slot into Unit 4 Lesson 9.5 or replace a workshop block).

Objective: Read and identify the 5 elements of a 1-page service agreement.

The 5 elements of any service agreement:

1. **Who** — both parties' names + contact.
2. **What** — what the service is, in specific language.
3. **When** — start date, end date, milestones.
4. **How much** — total fee, payment schedule, what counts as paid.
5. **What if** — what happens if either party can't deliver, and how to end the agreement.

Activity (15 min): Read a sample 1-page service agreement (provided in program assets). Identify each of the 5 elements. Highlight in 5 colors.

Reflection: "For your Capstone, write a 1-page service agreement template you'd use if you launched."

This half-lesson introduces real founder vocabulary without overwhelming. It's referenced again at Grade 11 (hiring) and Grade 12 (legal/ethical checklist).

ROUND 2 PATCH — Common Misconceptions (Grade 9)

Common kid-belief	Reality	Coaching move
"MVP means launch-ready."	MVP means minimum testable. The point is to learn faster, not launch faster. A paper prototype with 5 customer conversations is a real MVP.	Show three MVPs: (a) paper prototype, (b) one-feature website, (c) full product. Ask: "Which one teaches us the most for the lowest cost?" Often (a).
"If 5 people said yes, the BMC is validated."	Customer discovery is about patterns, not counts. 5 yeses can be 5 different reasons — fragile. 5 consistent yeses is signal.	Run a 5-quote analysis. Code each yes for the reason (job-to-be-done, frustration, willingness to pay). Inconsistent reasons = weak validation.
"P&L and cash flow are the same thing."	P&L (income statement) tracks revenue and expense over time. Cash flow tracks cash in and out over time. They differ on timing of receivables, prepayments, and depreciation.	Show a one-month example: revenue $10,000, profit $2,000, but cash on hand at month-end $500 (because customers havent paid). Point: profit ≠ cash.
"I dont need a contract for small jobs."	Even small jobs (under $500) benefit from a one-page service agreement — it prevents the next fight, not the lawsuit.	Show a service-agreement one-pager with 6 fields: parties, scope, price, payment terms, timeline, kill criteria. Walk through.
"If the BMC has all 9 boxes filled, the venture is real."	Boxes filled with evidence is real. Boxes filled with guesses is theater. The evidence column is the point.	Audit a sample BMC. Highlight which boxes have evidence (interview quote, count, observation) and which have guesses.

ROUND 2 PATCH — Worked Examples for One-Year P&L Lesson

> **“Most adults skip this step. You won't.”**
>
> - CASH, GRADE 9

Three fully worked P&Ls at different revenue scales.

P&L 1: Kid Lemonade (annualized). Revenue $500. COGS (lemons, sugar, cups) $200. Gross profit $300. Operating expenses (signage, refills, miscellaneous) $50. Net income $250. Gross margin = 60%. Net margin = 50%.

P&L 2: Teen Etsy Shop (annualized). Revenue $5,000. COGS (materials, packaging, shipping) $2,500. Gross profit $2,500. Operating expenses (listing fees, marketing, software) $750. Net income $1,750. Gross margin = 50%. Net margin = 35%.

P&L 3: Adult Side Business (one year). Revenue $50,000. COGS (services + materials) $20,000. Gross profit $30,000. Operating expenses (rent, software, contractor labor, marketing) $18,000. Net income $12,000. Gross margin = 60%. Net margin = 24%.

Coaching: "Note how net margin shrinks as the business grows — operating expenses scale faster than gross margin in the early years. The scaling story is in the operating-expense line."

Anti-example. Student writes: Revenue $5,000. Net income $5,000. Coaching: "Where did the COGS and op-ex go? Net income on $5K revenue is more often $500-$2,000."

ROUND 2 PATCH — Half-Lesson on Contracts Basics (Grade 9, NEW)

A 25-minute mini-lesson added to Lesson 9 (compliance survey). Goal: introduce the **service agreement** as a one-page template.

Service Agreement Template (one page)

```
 SERVICE AGREEMENT
Date: [DATE]

Parties:
- [PROVIDER NAME] (Provider)
- [CLIENT NAME] (Client)

1. Scope of work. Provider will: [SPECIFIC LIST OF DELIVERABLES].
2. Price. Client will pay: $[AMOUNT].
3. Payment terms. Payment due: [WHEN — e.g., on completion / 50% upfront, 50% on compl
4. Timeline. Work begins: [DATE]. Work completes: [DATE].
5. Changes. Any change to scope must be agreed in writing by both parties.
6. Termination. Either party can end this agreement with [NUMBER] days written notice.
7. Disputes. If we disagree, we will first try to resolve it directly. If we cant, [ST

Signed:
_______________________  [PROVIDER]  Date: _______
_______________________  [CLIENT]    Date: _______
```

Walk through each clause. Discuss what happens when each is missing. (Without scope: scope creep. Without termination: trapped. Without timeline: forever.) Have students fill in the template for their venture.

Note: this is not legal advice. For real business use, consult a lawyer. The skill is recognizing the shape of an agreement.

ROUND 2 PATCH — Multi-Class Branching Scenario (Grade 9)

"Two BMCs from last year — one ran a 2-week MVP test and killed, one iterated." A 2-day branching scenario.

- **Day 1.** Read both BMCs. The "killed" team had Customer Segment, Value Prop, and Channels but all assumptions in those boxes failed during MVP testing. The "iterated" team also had failed assumptions, but the team identified one assumption that didnt fail and pivoted to it. What was the assumption that broke?
- **Day 2.** Run a customer-discovery cycle on the iterated teams pivot — does it hold up? Write a kill/iterate/scale recommendation.

This branching practices BAVC (BMC), PSIG (5-interview discovery), MFL (P&L impact of pivot), CBM (positioning), OSO (functional decomposition), LTO (conflict-style — kill vs. iterate), ESL (compliance survey on the pivot), LRCI (MVP test).

ROUND 2 PATCH — Teacher Move Set (Grade 9)

- **Student insists "my MVP is launch-ready" after one paper prototype.** Say: "Whats the evidence it works? Show me the customer responses. If you dont have 5+ conversations, its not launch-ready — its a hypothesis."
- **Student fills BMC boxes with guesses.** Say: "Each box needs evidence. Whats the quote, count, or observation supporting this?" Audit the boxes one by one.
- **Student treats P&L and cash flow as the same.** Say: "Show me when each customer pays. If they pay 30 days later, your cash flow is delayed — even if the P&L looks fine."
- **Student writes a customer interview question that leads the witness** ("Do you love this idea?"). Say: "That question gets you the answer you want. Run the Mom Test version: What was the last time you faced this problem? What did you do?"
- **Student wants to skip the contracts lesson.** Say: "Even small jobs benefit from a one-page agreement. The lesson takes 25 minutes. Itll save you a fight in two years."

ENDNOTES & STANDARDS

Citations and alignment

Grade 9 cites 3-6 standards across CCSS, Jump$tart, CEE, CASEL, and C3 frameworks. Detailed alignment lives in the master Standards Alignment Map.

1. **Jump$tart Coalition Personal Finance Standards (2021 ed.).** Spending and Saving; Credit and Debt; Employment and Income; Investing; Risk Management and Insurance; Financial Decision Making.
2. **Council for Economic Education (CEE).** Voluntary National Content Standards in Economics.
3. **CCSS-ELA, Grades 11-12.** W.11-12.1, W.11-12.4, SL.11-12.1, SL.11-12.4, RST.11-12.7.
4. **CCSS-Math.** N-Q.1-3 (Reasoning with Quantities), S-IC (Statistics & Inferences), F-IF (Functions), MP.4 (Modeling).
5. **CASEL Core Competencies.** Self-Awareness, Self-Management, Social Awareness, Relationship Skills, Responsible Decision-Making.
6. **C3 Framework for Social Studies State Standards.** Dimension 2 (Economics, Civics) at the high-school grade band.
7. **NGSS-ETS Engineering Practices.** ETS1 (Engineering Design) for the test-and-iterate spine of the year.
8. **Cash to Company Master Scope & Sequence.** 13 grades x 8 strands x 36 weeks; outcomes quoted verbatim from _master/00_Master_Scope_and_Sequence.md.
9. **Cash Mascot Style Guide.** 9-12 voice rules: multi-sentence arguments; vocabulary includes unit economics, runway, premortem, MVP; never flatters, never promises wealth.
10. **Sensitivity protocol.** No probing of family finances; financial scenarios framed as hypothetical for a sample household, business, or scenario.

One promise.

Cash to Company makes one promise: students who move t
program will be clearer thinkers about money and entrepren
matter what they end up doing for a living.

We do not promise wealth. We do not promise success. We pro

> **“From flash of inspiration to repeatable discipline.”**

- CASH, GRADE 9

For educational use. This curriculum is informational and does not constitute financial, legal, or tax advice. Standards alignment cited per the Cash to Company Standards Alignment Map (Jump$tart, CEE, CCSS, C3, CASEL, NGSS-ETS).